Variable Selection and Classification

1.

Consider the features and response that you identified or created in the previous HW. Use variable selection to choose a reduced model that offers a reasonable balance between predictive ability and interpretation.

```r
library(fastDummies)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
# data
hours_clean = read.csv("../data/hour_clean.csv")
hours = hours_clean[, c("yr", "mnth", "workingday", "weathersit", "atemp", "hum", "windspeed", "cnt")]
hours$weathersit[hours$weathersit == 4] = 3
hours = dummy_cols(hours,
                   select_columns = c("mnth", "weathersit"),
                   remove_first_dummy = TRUE,
                   remove_selected_columns = TRUE)
hours = dplyr::select(hours, -cnt, cnt)
head(hours)
```

```
##   yr workingday  atemp  hum windspeed mnth_2 mnth_3 mnth_4 mnth_5 mnth_6 mnth_7
## 1  0          0 0.2879 0.81    0.0000      0      0      0      0      0      0
## 2  0          0 0.2727 0.80    0.0000      0      0      0      0      0      0
## 3  0          0 0.2727 0.80    0.0000      0      0      0      0      0      0
## 4  0          0 0.2879 0.75    0.0000      0      0      0      0      0      0
## 5  0          0 0.2879 0.75    0.0000      0      0      0      0      0      0
## 6  0          0 0.2576 0.75    0.0896      0      0      0      0      0      0
##   mnth_8 mnth_9 mnth_10 mnth_11 mnth_12 weathersit_2 weathersit_3 cnt
## 1      0      0       0       0       0            0            0  16
## 2      0      0       0       0       0            0            0  40
## 3      0      0       0       0       0            0            0  32
## 4      0      0       0       0       0            0            0  13
## 5      0      0       0       0       0            0            0   1
## 6      0      0       0       0       0            1            0   1
```

```r
# current model

model = lm(sqrt(cnt) ~ poly(atemp, 3) * poly(hum, 3) + poly(windspeed, 2) + . - atemp - hum, data = hou
summary(model)
```

```
##
## Call:
## lm(formula = sqrt(cnt) ~ poly(atemp, 3) * poly(hum, 3) + poly(windspeed,
##     2) + . - atemp - hum, data = hours)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -19.373  -3.702  -0.189   3.333  20.484
##
## Coefficients: (1 not defined because of singularities)
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   1.144e+01  2.095e-01  54.577  < 2e-16 ***
## poly(atemp, 3)1               4.721e+02  1.346e+01  35.074  < 2e-16 ***
## poly(atemp, 3)2              -1.453e+00  9.759e+00  -0.149 0.881618
## poly(atemp, 3)3              -6.475e+01  9.888e+00  -6.549 5.97e-11 ***
## poly(hum, 3)1                -3.186e+02  9.425e+00 -33.807  < 2e-16 ***
## poly(hum, 3)2                -2.780e+01  7.593e+00  -3.661 0.000252 ***
## poly(hum, 3)3                 1.466e+01  8.108e+00   1.807 0.070704 .
## poly(windspeed, 2)1           3.839e+01  5.799e+00   6.621 3.68e-11 ***
## poly(windspeed, 2)2          -3.492e+01  5.378e+00  -6.492 8.68e-11 ***
## yr                            2.291e+00  8.192e-02  27.971  < 2e-16 ***
## workingday                    6.441e-02  8.742e-02   0.737 0.461280
## windspeed                            NA         NA      NA       NA
## mnth_2                       -4.075e-01  2.110e-01  -1.931 0.053532 .
## mnth_3                       -4.060e-01  2.184e-01  -1.859 0.063055 .
## mnth_4                       -1.205e+00  2.355e-01  -5.117 3.13e-07 ***
## mnth_5                       -8.502e-01  2.598e-01  -3.273 0.001066 **
## mnth_6                       -3.353e+00  2.757e-01 -12.163  < 2e-16 ***
## mnth_7                       -4.239e+00  2.975e-01 -14.251  < 2e-16 ***
## mnth_8                       -2.816e+00  2.830e-01  -9.949  < 2e-16 ***
## mnth_9                       -2.428e-01  2.675e-01  -0.908 0.364080
## mnth_10                       1.147e+00  2.424e-01   4.734 2.22e-06 ***
## mnth_11                       1.466e+00  2.222e-01   6.598 4.28e-11 ***
## mnth_12                       1.340e+00  2.139e-01   6.266 3.80e-10 ***
## weathersit_2                  8.053e-01  9.902e-02   8.133 4.47e-16 ***
## weathersit_3                  8.418e-03  1.719e-01   0.049 0.960934
## poly(atemp, 3)1:poly(hum, 3)1 -1.338e+04  1.670e+03  -8.013 1.19e-15 ***
## poly(atemp, 3)2:poly(hum, 3)1  2.315e+03  1.475e+03   1.570 0.116425
## poly(atemp, 3)3:poly(hum, 3)1  2.504e+03  1.521e+03   1.647 0.099626 .
## poly(atemp, 3)1:poly(hum, 3)2 -6.961e+03  1.335e+03  -5.213 1.88e-07 ***
## poly(atemp, 3)2:poly(hum, 3)2 -4.697e+01  1.184e+03  -0.040 0.968362
## poly(atemp, 3)3:poly(hum, 3)2 -2.227e+03  1.201e+03  -1.854 0.063749 .
## poly(atemp, 3)1:poly(hum, 3)3 -2.441e+03  1.383e+03  -1.765 0.077619 .
## poly(atemp, 3)2:poly(hum, 3)3 -5.199e+03  1.160e+03  -4.484 7.39e-06 ***
## poly(atemp, 3)3:poly(hum, 3)3  2.797e+03  1.239e+03   2.257 0.024012 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.302 on 17245 degrees of freedom
## Multiple R-squared:  0.3786, Adjusted R-squared:  0.3775
## F-statistic: 328.4 on 32 and 17245 DF,  p-value: < 2.2e-16
```

```r
# variable selection with AIC
step_aic = step(model, direction = "both")
```

```
## Start:  AIC=57673.28
## sqrt(cnt) ~ poly(atemp, 3) * poly(hum, 3) + poly(windspeed, 2) +
##     (yr + workingday + atemp + hum + windspeed + mnth_2 + mnth_3 +
##         mnth_4 + mnth_5 + mnth_6 + mnth_7 + mnth_8 + mnth_9 +
##         mnth_10 + mnth_11 + mnth_12 + weathersit_2 + weathersit_3) -
##     atemp - hum
##
##
## Step:  AIC=57673.28
## sqrt(cnt) ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed, 2) +
##     yr + workingday + mnth_2 + mnth_3 + mnth_4 + mnth_5 + mnth_6 +
##     mnth_7 + mnth_8 + mnth_9 + mnth_10 + mnth_11 + mnth_12 +
##     weathersit_2 + weathersit_3 + poly(atemp, 3):poly(hum, 3)
##
##                               Df Sum of Sq    RSS   AIC
## - weathersit_3                 1       0.1 484721 57671
## - workingday                   1      15.3 484737 57672
## - mnth_9                       1      23.2 484745 57672
## <none>                                     484721 57673
## - mnth_3                       1      97.1 484819 57675
## - mnth_2                       1     104.8 484826 57675
## - mnth_5                       1     301.1 485023 57682
## - mnth_10                      1     630.0 485351 57694
## - mnth_4                       1     736.1 485458 57697
## - mnth_12                      1    1103.5 485825 57711
## - mnth_11                      1    1223.8 485945 57715
## - weathersit_2                 1    1859.2 486581 57737
## - poly(windspeed, 2)           2    2457.8 487179 57757
## - mnth_8                       1    2782.5 487504 57770
## - mnth_6                       1    4158.3 488880 57819
## - mnth_7                       1    5708.1 490430 57874
## - poly(atemp, 3):poly(hum, 3)  9    9540.0 494261 57992
## - yr                           1   21990.3 506712 58438
##
## Step:  AIC=57671.28
## sqrt(cnt) ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed, 2) +
##     yr + workingday + mnth_2 + mnth_3 + mnth_4 + mnth_5 + mnth_6 +
##     mnth_7 + mnth_8 + mnth_9 + mnth_10 + mnth_11 + mnth_12 +
##     weathersit_2 + poly(atemp, 3):poly(hum, 3)
##
##                               Df Sum of Sq    RSS   AIC
## - workingday                   1      15.4 484737 57670
## - mnth_9                       1      23.2 484745 57670
## <none>                                     484721 57671
## - mnth_3                       1      97.3 484819 57673
## - mnth_2                       1     104.7 484826 57673
## + weathersit_3                 1       0.1 484721 57673
## - mnth_5                       1     301.4 485023 57680
## - mnth_10                      1     629.9 485351 57692
## - mnth_4                       1     736.1 485458 57695
## - mnth_12                      1    1103.4 485825 57709
## - mnth_11                      1    1223.7 485945 57713
## - weathersit_2                 1    2009.7 486731 57741
## - poly(windspeed, 2)           2    2503.8 487225 57756
```

```
## - mnth_8                        1    2783.5 487505 57768
## - mnth_6                        1    4159.1 488881 57817
## - mnth_7                        1    5709.9 490431 57872
## - poly(atemp, 3):poly(hum, 3)  9    9585.6 494307 57992
## - yr                            1   22005.5 506727 58436
##
## Step:  AIC=57669.83
## sqrt(cnt) ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed, 2) +
##     yr + mnth_2 + mnth_3 + mnth_4 + mnth_5 + mnth_6 + mnth_7 +
##     mnth_8 + mnth_9 + mnth_10 + mnth_11 + mnth_12 + weathersit_2 +
##     poly(atemp, 3):poly(hum, 3)
##
##                               Df Sum of Sq    RSS   AIC
## - mnth_9                        1      24.5 484761 57669
## <none>                                      484737 57670
## - mnth_3                        1      96.0 484833 57671
## + workingday                    1      15.4 484721 57671
## - mnth_2                        1     103.8 484841 57672
## + weathersit_3                  1       0.2 484737 57672
## - mnth_5                        1     304.5 485041 57679
## - mnth_10                       1     625.5 485362 57690
## - mnth_4                        1     740.8 485478 57694
## - mnth_12                       1    1101.8 485839 57707
## - mnth_11                       1    1221.3 485958 57711
## - weathersit_2                  1    2018.1 486755 57740
## - poly(windspeed, 2)           2    2502.5 487239 57755
## - mnth_8                        1    2782.7 487520 57767
## - mnth_6                        1    4167.2 488904 57816
## - mnth_7                        1    5732.6 490469 57871
## - poly(atemp, 3):poly(hum, 3)  9    9579.7 494317 57990
## - yr                            1   21997.8 506735 58435
##
## Step:  AIC=57668.7
## sqrt(cnt) ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed, 2) +
##     yr + mnth_2 + mnth_3 + mnth_4 + mnth_5 + mnth_6 + mnth_7 +
##     mnth_8 + mnth_10 + mnth_11 + mnth_12 + weathersit_2 + poly(atemp,
##     3):poly(hum, 3)
##
##                               Df Sum of Sq    RSS   AIC
## <none>                                      484761 57669
## - mnth_3                        1      71.9 484833 57669
## - mnth_2                        1      79.6 484841 57670
## + mnth_9                        1      24.5 484737 57670
## + workingday                    1      16.6 484745 57670
## + weathersit_3                  1       0.3 484761 57671
## - mnth_5                        1     398.9 485160 57681
## - mnth_4                        1     941.5 485703 57700
## - mnth_10                       1    1417.9 486179 57717
## - mnth_12                       1    1680.1 486441 57726
## - mnth_11                       1    1986.8 486748 57737
## - weathersit_2                  1    2058.7 486820 57740
## - poly(windspeed, 2)           2    2503.5 487265 57754
## - mnth_8                        1    5049.0 489810 57846
## - mnth_6                        1    7663.4 492425 57938
```

```
## - poly(atemp, 3):poly(hum, 3)  9    9768.2 494529 57995
## - mnth_7                       1   10340.1 495101 58031
## - yr                           1   22146.0 506907 58439
```

```r
summary(step_aic)
```

```
##
## Call:
## lm(formula = sqrt(cnt) ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed,
##     2) + yr + mnth_2 + mnth_3 + mnth_4 + mnth_5 + mnth_6 + mnth_7 +
##     mnth_8 + mnth_10 + mnth_11 + mnth_12 + weathersit_2 + poly(atemp,
##     3):poly(hum, 3), data = hours)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.3175  -3.6993  -0.1836   3.3265  20.3892
##
## Coefficients:
##                                Estimate Std. Error t value Pr(>|t|)
## (Intercept)                   1.134e+01  1.234e-01  91.844  < 2e-16 ***
## poly(atemp, 3)1               4.663e+02  1.168e+01  39.908  < 2e-16 ***
## poly(atemp, 3)2              -8.366e-01  9.713e+00  -0.086 0.931364
## poly(atemp, 3)3              -6.401e+01  9.841e+00  -6.504 8.01e-11 ***
## poly(hum, 3)1                -3.199e+02  8.987e+00 -35.593  < 2e-16 ***
## poly(hum, 3)2                -2.773e+01  7.512e+00  -3.691 0.000224 ***
## poly(hum, 3)3                 1.447e+01  8.102e+00   1.786 0.074134 .
## poly(windspeed, 2)1           3.851e+01  5.718e+00   6.735 1.69e-11 ***
## poly(windspeed, 2)2          -3.484e+01  5.377e+00  -6.480 9.42e-11 ***
## yr                            2.295e+00  8.177e-02  28.071  < 2e-16 ***
## mnth_2                       -3.180e-01  1.890e-01  -1.683 0.092441 .
## mnth_3                       -2.906e-01  1.817e-01  -1.599 0.109730
## mnth_4                       -1.073e+00  1.854e-01  -5.788 7.26e-09 ***
## mnth_5                       -6.807e-01  1.807e-01  -3.767 0.000166 ***
## mnth_6                       -3.172e+00  1.921e-01 -16.513  < 2e-16 ***
## mnth_7                       -4.050e+00  2.112e-01 -19.181  < 2e-16 ***
## mnth_8                       -2.625e+00  1.959e-01 -13.403  < 2e-16 ***
## mnth_10                       1.292e+00  1.819e-01   7.103 1.27e-12 ***
## mnth_11                       1.576e+00  1.874e-01   8.408  < 2e-16 ***
## mnth_12                       1.438e+00  1.859e-01   7.732 1.12e-14 ***
## weathersit_2                  8.116e-01  9.482e-02   8.559  < 2e-16 ***
## poly(atemp, 3)1:poly(hum, 3)1 -1.345e+04  1.664e+03  -8.081 6.85e-16 ***
## poly(atemp, 3)2:poly(hum, 3)1  2.286e+03  1.474e+03   1.551 0.121034
## poly(atemp, 3)3:poly(hum, 3)1  2.582e+03  1.515e+03   1.704 0.088328 .
## poly(atemp, 3)1:poly(hum, 3)2 -6.993e+03  1.332e+03  -5.251 1.53e-07 ***
## poly(atemp, 3)2:poly(hum, 3)2 -1.239e+02  1.182e+03  -0.105 0.916553
## poly(atemp, 3)3:poly(hum, 3)2 -2.197e+03  1.199e+03  -1.833 0.066870 .
## poly(atemp, 3)1:poly(hum, 3)3 -2.464e+03  1.382e+03  -1.783 0.074672 .
## poly(atemp, 3)2:poly(hum, 3)3 -5.239e+03  1.159e+03  -4.520 6.23e-06 ***
## poly(atemp, 3)3:poly(hum, 3)3  2.847e+03  1.234e+03   2.307 0.021077 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.301 on 17248 degrees of freedom
## Multiple R-squared:  0.3786, Adjusted R-squared:  0.3775
## F-statistic: 362.4 on 29 and 17248 DF,  p-value: < 2.2e-16
```

```r
# variable selection with BIC

n = nrow(hours)
step_bic = step(model, direction = "both", k = log(n))
```

```
## Start:  AIC=57929.26
## sqrt(cnt) ~ poly(atemp, 3) * poly(hum, 3) + poly(windspeed, 2) +
##     (yr + workingday + atemp + hum + windspeed + mnth_2 + mnth_3 +
##         mnth_4 + mnth_5 + mnth_6 + mnth_7 + mnth_8 + mnth_9 +
##         mnth_10 + mnth_11 + mnth_12 + weathersit_2 + weathersit_3) -
##     atemp - hum
##
##
## Step:  AIC=57929.26
## sqrt(cnt) ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed, 2) +
##     yr + workingday + mnth_2 + mnth_3 + mnth_4 + mnth_5 + mnth_6 +
##     mnth_7 + mnth_8 + mnth_9 + mnth_10 + mnth_11 + mnth_12 +
##     weathersit_2 + weathersit_3 + poly(atemp, 3):poly(hum, 3)
##
##                               Df Sum of Sq    RSS   AIC
## - weathersit_3                 1       0.1 484721 57920
## - workingday                   1      15.3 484737 57920
## - mnth_9                       1      23.2 484745 57920
## - mnth_3                       1      97.1 484819 57923
## - mnth_2                       1     104.8 484826 57923
## <none>                                     484721 57929
## - mnth_5                       1     301.1 485023 57930
## - mnth_10                      1     630.0 485351 57942
## - mnth_4                       1     736.1 485458 57946
## - mnth_12                      1    1103.5 485825 57959
## - mnth_11                      1    1223.8 485945 57963
## - weathersit_2                 1    1859.2 486581 57986
## - poly(windspeed, 2)           2    2457.8 487179 57997
## - mnth_8                       1    2782.5 487504 58018
## - mnth_6                       1    4158.3 488880 58067
## - mnth_7                       1    5708.1 490430 58122
## - poly(atemp, 3):poly(hum, 3)  9    9540.0 494261 58178
## - yr                           1   21990.3 506712 58686
##
## Step:  AIC=57919.51
## sqrt(cnt) ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed, 2) +
##     yr + workingday + mnth_2 + mnth_3 + mnth_4 + mnth_5 + mnth_6 +
##     mnth_7 + mnth_8 + mnth_9 + mnth_10 + mnth_11 + mnth_12 +
##     weathersit_2 + poly(atemp, 3):poly(hum, 3)
##
##                               Df Sum of Sq    RSS   AIC
## - workingday                   1      15.4 484737 57910
## - mnth_9                       1      23.2 484745 57911
## - mnth_3                       1      97.3 484819 57913
## - mnth_2                       1     104.7 484826 57913
## <none>                                     484721 57920
## - mnth_5                       1     301.4 485023 57920
## + weathersit_3                 1       0.1 484721 57929
## - mnth_10                      1     629.9 485351 57932
```

```
## - mnth_4                        1     736.1 485458 57936
## - mnth_12                       1    1103.4 485825 57949
## - mnth_11                       1    1223.7 485945 57953
## - weathersit_2                  1    2009.7 486731 57981
## - poly(windspeed, 2)            2    2503.8 487225 57989
## - mnth_8                        1    2783.5 487505 58009
## - mnth_6                        1    4159.1 488881 58057
## - mnth_7                        1    5709.9 490431 58112
## - poly(atemp, 3):poly(hum, 3)  9    9585.6 494307 58170
## - yr                            1   22005.5 506727 58677
##
## Step:  AIC=57910.3
## sqrt(cnt) ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed, 2) +
##     yr + mnth_2 + mnth_3 + mnth_4 + mnth_5 + mnth_6 + mnth_7 +
##     mnth_8 + mnth_9 + mnth_10 + mnth_11 + mnth_12 + weathersit_2 +
##     poly(atemp, 3):poly(hum, 3)
##
##                               Df Sum of Sq    RSS   AIC
## - mnth_9                        1      24.5 484761 57901
## - mnth_3                        1      96.0 484833 57904
## - mnth_2                        1     103.8 484841 57904
## <none>                                     484737 57910
## - mnth_5                        1     304.5 485041 57911
## + workingday                    1      15.4 484721 57920
## + weathersit_3                  1       0.2 484737 57920
## - mnth_10                       1     625.5 485362 57923
## - mnth_4                        1     740.8 485478 57927
## - mnth_12                       1    1101.8 485839 57940
## - mnth_11                       1    1221.3 485958 57944
## - weathersit_2                  1    2018.1 486755 57972
## - poly(windspeed, 2)            2    2502.5 487239 57980
## - mnth_8                        1    2782.7 487520 57999
## - mnth_6                        1    4167.2 488904 58048
## - mnth_7                        1    5732.6 490469 58104
## - poly(atemp, 3):poly(hum, 3)  9    9579.7 494317 58161
## - yr                            1   21997.8 506735 58667
##
## Step:  AIC=57901.41
## sqrt(cnt) ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed, 2) +
##     yr + mnth_2 + mnth_3 + mnth_4 + mnth_5 + mnth_6 + mnth_7 +
##     mnth_8 + mnth_10 + mnth_11 + mnth_12 + weathersit_2 + poly(atemp,
##     3):poly(hum, 3)
##
##                               Df Sum of Sq    RSS   AIC
## - mnth_3                        1      71.9 484833 57894
## - mnth_2                        1      79.6 484841 57894
## <none>                                     484761 57901
## - mnth_5                        1     398.9 485160 57906
## + mnth_9                        1      24.5 484737 57910
## + workingday                    1      16.6 484745 57911
## + weathersit_3                  1       0.3 484761 57911
## - mnth_4                        1     941.5 485703 57925
## - mnth_10                       1    1417.9 486179 57942
## - mnth_12                       1    1680.1 486441 57951
```

```
## - mnth_11                      1    1986.8 486748 57962
## - weathersit_2                 1    2058.7 486820 57965
## - poly(windspeed, 2)           2    2503.5 487265 57971
## - mnth_8                       1    5049.0 489810 58071
## - poly(atemp, 3):poly(hum, 3)  9    9768.2 494529 58158
## - mnth_6                       1    7663.4 492425 58163
## - mnth_7                       1   10340.1 495101 58256
## - yr                           1   22146.0 506907 58663
##
## Step:  AIC=57894.22
## sqrt(cnt) ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed, 2) +
##     yr + mnth_2 + mnth_4 + mnth_5 + mnth_6 + mnth_7 + mnth_8 +
##     mnth_10 + mnth_11 + mnth_12 + weathersit_2 + poly(atemp,
##     3):poly(hum, 3)
##
##                               Df Sum of Sq    RSS   AIC
## - mnth_2                       1      37.2 484870 57886
## <none>                                     484833 57894
## - mnth_5                       1     334.0 485167 57896
## + mnth_3                       1      71.9 484761 57901
## + workingday                   1      13.9 484819 57903
## + mnth_9                       1       0.3 484833 57904
## + weathersit_3                 1       0.3 484833 57904
## - mnth_4                       1     887.1 485720 57916
## - mnth_10                      1    1915.3 486749 57953
## - weathersit_2                 1    2045.1 486878 57957
## - poly(windspeed, 2)           2    2458.1 487291 57962
## - mnth_12                      1    2360.0 487193 57968
## - mnth_11                      1    2778.7 487612 57983
## - mnth_8                       1    5019.0 489852 58062
## - poly(atemp, 3):poly(hum, 3)  9    9696.3 494530 58149
## - mnth_6                       1    7766.0 492599 58159
## - mnth_7                       1   10371.5 495205 58250
## - yr                           1   22208.9 507042 58658
##
## Step:  AIC=57885.79
## sqrt(cnt) ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed, 2) +
##     yr + mnth_4 + mnth_5 + mnth_6 + mnth_7 + mnth_8 + mnth_10 +
##     mnth_11 + mnth_12 + weathersit_2 + poly(atemp, 3):poly(hum,
##     3)
##
##                               Df Sum of Sq    RSS   AIC
## <none>                                     484870 57886
## - mnth_5                       1     311.9 485182 57887
## + mnth_2                       1      37.2 484833 57894
## + mnth_3                       1      29.5 484841 57894
## + workingday                   1      13.3 484857 57895
## + mnth_9                       1       3.4 484867 57895
## + weathersit_3                 1       0.2 484870 57896
## - mnth_4                       1     850.5 485721 57906
## - weathersit_2                 1    2044.9 486915 57949
## - mnth_10                      1    2096.7 486967 57951
## - poly(windspeed, 2)           2    2484.2 487355 57955
## - mnth_12                      1    2775.0 487645 57975
```

```
## - mnth_11                       1    3199.9 488070 57990
## - mnth_8                        1    4982.2 489853 58053
## - poly(atemp, 3):poly(hum, 3)  9    9659.8 494530 58139
## - mnth_6                        1    7733.4 492604 58149
## - mnth_7                        1   10334.7 495205 58240
## - yr                            1   22200.8 507071 58650
```

```r
summary(step_bic)
```

```
##
## Call:
## lm(formula = sqrt(cnt) ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed,
##     2) + yr + mnth_4 + mnth_5 + mnth_6 + mnth_7 + mnth_8 + mnth_10 +
##     mnth_11 + mnth_12 + weathersit_2 + poly(atemp, 3):poly(hum,
##     3), data = hours)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.4699  -3.7039  -0.1875   3.3242  20.4719
##
## Coefficients:
##                                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)                    1.120e+01  1.017e-01 110.172  < 2e-16 ***
## poly(atemp, 3)1                4.705e+02  1.146e+01  41.048  < 2e-16 ***
## poly(atemp, 3)2                1.443e+00  9.606e+00   0.150 0.880565
## poly(atemp, 3)3               -6.763e+01  9.667e+00  -6.996 2.73e-12 ***
## poly(hum, 3)1                 -3.183e+02  8.951e+00 -35.561  < 2e-16 ***
## poly(hum, 3)2                 -2.824e+01  7.508e+00  -3.761 0.000170 ***
## poly(hum, 3)3                  1.468e+01  8.099e+00   1.813 0.069892 .
## poly(windspeed, 2)1            3.764e+01  5.698e+00   6.605 4.08e-11 ***
## poly(windspeed, 2)2           -3.520e+01  5.363e+00  -6.563 5.42e-11 ***
## yr                             2.298e+00  8.176e-02  28.104  < 2e-16 ***
## mnth_4                        -9.091e-01  1.653e-01  -5.501 3.84e-08 ***
## mnth_5                        -5.703e-01  1.712e-01  -3.331 0.000866 ***
## mnth_6                        -3.077e+00  1.855e-01 -16.587  < 2e-16 ***
## mnth_7                        -3.973e+00  2.072e-01 -19.175  < 2e-16 ***
## mnth_8                        -2.543e+00  1.910e-01 -13.313  < 2e-16 ***
## mnth_10                        1.436e+00  1.663e-01   8.637  < 2e-16 ***
## mnth_11                        1.753e+00  1.643e-01  10.670  < 2e-16 ***
## mnth_12                        1.615e+00  1.626e-01   9.936  < 2e-16 ***
## weathersit_2                   8.087e-01  9.481e-02   8.529  < 2e-16 ***
## poly(atemp, 3)1:poly(hum, 3)1 -1.338e+04  1.664e+03  -8.043 9.37e-16 ***
## poly(atemp, 3)2:poly(hum, 3)1  2.184e+03  1.472e+03   1.483 0.138078
## poly(atemp, 3)3:poly(hum, 3)1  2.416e+03  1.513e+03   1.597 0.110232
## poly(atemp, 3)1:poly(hum, 3)2 -6.943e+03  1.331e+03  -5.217 1.84e-07 ***
## poly(atemp, 3)2:poly(hum, 3)2 -9.135e+01  1.182e+03  -0.077 0.938404
## poly(atemp, 3)3:poly(hum, 3)2 -2.319e+03  1.197e+03  -1.938 0.052687 .
## poly(atemp, 3)1:poly(hum, 3)3 -2.437e+03  1.382e+03  -1.763 0.077875 .
## poly(atemp, 3)2:poly(hum, 3)3 -5.243e+03  1.159e+03  -4.524 6.12e-06 ***
## poly(atemp, 3)3:poly(hum, 3)3  2.821e+03  1.234e+03   2.286 0.022267 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.302 on 17250 degrees of freedom
## Multiple R-squared:  0.3785, Adjusted R-squared:  0.3775
```

```
## F-statistic:   389 on 27 and 17250 DF,  p-value: < 2.2e-16
```

After running stepwise variable selection with both AIC and BIC, we selected a reduced model using BIC, which retained 27 predictors. The model explains approximately 37.75% of the variance in the square root of hourly bike counts. Key predictors include atemp, hum, windspeed, month, year, and their interactions. Notably, ridership increased substantially from 2011 to 2012 (yr coefficient = 2.298, $p < 0.001$) and decreased during summer months (mnth_6, mnth_7, mnth_8) likely due to heat or vacations. We also observe strong nonlinear interactions between perceived temperature and humidity, such that extremely hot and humid conditions drastically reduce ridership (poly(atemp, 3)1:poly(hum, 3)1 = –13,380, $p < 0.001$).

2

```r
# binarize response variable
threshold = median(hours$cnt)
hours$cnt_bin = as.integer(hours$cnt > threshold)
```

```r
log_reg = glm(cnt_bin ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed,
    2) + yr + mnth_4 + mnth_5 + mnth_6 + mnth_7 + mnth_8 + mnth_10 +
    mnth_11 + mnth_12 + weathersit_2 + poly(atemp, 3):poly(hum,
    3), family = binomial, data = hours)
summary(log_reg)
```

```
##
## Call:
## glm(formula = cnt_bin ~ poly(atemp, 3) + poly(hum, 3) + poly(windspeed,
##     2) + yr + mnth_4 + mnth_5 + mnth_6 + mnth_7 + mnth_8 + mnth_10 +
##     mnth_11 + mnth_12 + weathersit_2 + poly(atemp, 3):poly(hum,
##     3), family = binomial, data = hours)
##
## Coefficients:
##                                 Estimate Std. Error z value Pr(>|z|)
## (Intercept)                   -2.724e-01  4.881e-02  -5.582 2.38e-08 ***
## poly(atemp, 3)1                1.897e+02  5.839e+00  32.486  < 2e-16 ***
## poly(atemp, 3)2                5.597e+00  4.613e+00   1.213 0.225052
## poly(atemp, 3)3               -2.394e+01  4.845e+00  -4.941 7.78e-07 ***
## poly(hum, 3)1                 -1.222e+02  4.716e+00 -25.911  < 2e-16 ***
## poly(hum, 3)2                 -6.806e+00  3.727e+00  -1.826 0.067852 .
## poly(hum, 3)3                  3.514e+00  4.871e+00   0.721 0.470639
## poly(windspeed, 2)1            1.203e+01  2.652e+00   4.535 5.76e-06 ***
## poly(windspeed, 2)2           -8.299e+00  2.485e+00  -3.339 0.000840 ***
## yr                             7.790e-01  3.806e-02  20.469  < 2e-16 ***
## mnth_4                        -3.020e-01  7.845e-02  -3.850 0.000118 ***
## mnth_5                        -7.568e-02  8.029e-02  -0.943 0.345877
## mnth_6                        -1.150e+00  8.982e-02 -12.802  < 2e-16 ***
## mnth_7                        -1.622e+00  1.012e-01 -16.039  < 2e-16 ***
## mnth_8                        -9.713e-01  9.190e-02 -10.569  < 2e-16 ***
## mnth_10                        5.658e-01  7.589e-02   7.455 8.98e-14 ***
## mnth_11                        8.659e-01  7.250e-02  11.944  < 2e-16 ***
## mnth_12                        7.891e-01  7.147e-02  11.040  < 2e-16 ***
## weathersit_2                   3.820e-01  4.309e-02   8.866  < 2e-16 ***
## poly(atemp, 3)1:poly(hum, 3)1 -5.208e+03  9.094e+02  -5.727 1.02e-08 ***
## poly(atemp, 3)2:poly(hum, 3)1 -9.070e+01  7.696e+02  -0.118 0.906184
## poly(atemp, 3)3:poly(hum, 3)1 -6.774e+01  8.443e+02  -0.080 0.936054
## poly(atemp, 3)1:poly(hum, 3)2 -1.391e+03  6.758e+02  -2.058 0.039636 *
## poly(atemp, 3)2:poly(hum, 3)2  7.168e+02  5.914e+02   1.212 0.225473
## poly(atemp, 3)3:poly(hum, 3)2  1.448e+02  6.644e+02   0.218 0.827467
## poly(atemp, 3)1:poly(hum, 3)3 -7.324e+02  8.908e+02  -0.822 0.410928
## poly(atemp, 3)2:poly(hum, 3)3 -2.892e+03  7.294e+02  -3.965 7.33e-05 ***
## poly(atemp, 3)3:poly(hum, 3)3  6.890e+02  7.640e+02   0.902 0.367130
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
```

```
##     Null deviance: 23952  on 17277  degrees of freedom
## Residual deviance: 17371  on 17250  degrees of freedom
## AIC: 17427
##
## Number of Fisher Scoring iterations: 6
```

```r
# compute accuracy

probs = predict(log_reg, type = "response")
preds = ifelse(probs > 0.5, 1, 0)
accuracy = mean(preds == hours$cnt_bin)
print(paste("Accuracy:", accuracy))
```

```
## [1] "Accuracy: 0.752170390091446"
```

To binarize the `cnt` (response) column, we used the median as the threashold. We trained a logistic regression model using the same predictors from our final linear model for binary classification. The model achieved an accuracy of 75.22%, indicating strong predictive performance. Temperature, humidity, windspeed, seasonality, and weather conditions were all significant predictors of high demand. In particular, high temperature increased the odds of high demand, but extreme heat combined with high humidity drastically reduced it. The model also captured strong seasonal effects and a marked increase in demand in 2012 compared to 2011.

Contribution Statement

Members: Ashley Ho & Mizuho Fukuda

1. We implemented and interpreted the results of both variable selection methods together. We took turns coding and writing.
2. We implemented and interpreted the logistic regression together. We took turns coding and writing.